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AI for Bharat Hackathon

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Team Name : KisaanMitra.AI

Team Leader Name : Aditya Mahesh Rane

Problem Statement : AI for Rural Innovation & Sustainable Systems - Build an AI-powered solution that supports rural ecosystems, sustainability, or resource-efficient systems.

Brief about the Idea:

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Problem Statement

- India has **146.5 million** farming households, of which **86% are small and marginal farmers** owning less than 2 hectares *(Agriculture Census 2015-16)*. **Over 50% of these households are in debt** *(NSO 2019)*, with an average monthly income of just ₹10,218 (NSS 77th Round).
- They lack access to timely, hyper-local and integrated decision support across **crop management, financial planning** and **market intelligence** — leading to annual post-harvest and pre-harvest losses worth **₹92,651 crore** *(ICAR-CIPHET),* with fruits and vegetables alone losing 5-15% of produce before reaching the market (NABCONS 2022).
- **Existing information is fragmented**, non-personalized and difficult to apply at the village level. 80% of marginal farmers are affected by adverse climatic events *(FEED 2024),* yet most digital agri-solutions require smartphones, English literacy, and app downloads — excluding the very farmers who need help the most.
- **Who is Affected -** Small and marginal farmers comprising **86%** of India's agricultural households *(Agriculture Census 2015-16),* with 65% being marginal farmers whose operational holdings cover only 24% of total cultivable land. They face a ₹50,000+ annual income gap due to poor market timing, delayed disease detection, and lack of access to credit and government schemes.

Our Solution

KisaanMitra.AI is a WhatsApp-based multi-agent AI system designed to act as a farmer's all-in-one intelligent assistant.

It combines three specialized AI agents:

- **Crop Agent –** Provides hyper-local crop guidance, disease diagnosis, fertilizer and pesticide recommendations and season-specific best practices tailored to the farmer's village and crop stage.
- **Market Agent –** Uses demand trends and time-series forecasting to suggest the best crop to grow, optimal harvest timing, and where to sell for maximum profit.
- **Finance Agent –** Provides end-to-end budget planning for farmers. Helps farmers plan pre-planting budgets, optimize input costs, access relevant government schemes and make informed financial decisions.

The solution also has **local language support**. It operates in Hindi and regional languages with simple, conversational interaction. Supports text and image inputs to ensure accessibility for farmers with limited digital experience.

Together, these agents deliver personalized, data-driven decisions directly through WhatsApp making advanced agricultural intelligence simple, accessible and actionable for every farmer

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Why AI is Required

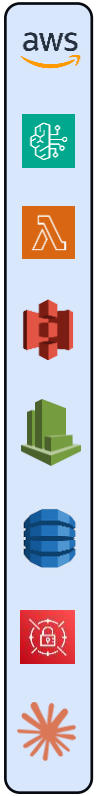
- Farmers speak natural Hindi/Hinglish with typos and slang — only an LLM can **understand intent, detect language, and route to the right agent (Crop/Market/Finance)** from a single message
- Disease diagnosis from crop photos needs **multi-modal AI** to identify 50+ diseases at 85%+ accuracy and generate treatment + cost estimates instantly
- Personalized budgets, price forecasts, and scheme eligibility require multi-step reasoning across each farmer's unique profile, location, and real-time data — **impossible with rule-based systems**

How AWS Services Are Used

- **Amazon Bedrock (Nova Pro + Claude Sonnet 4):** Core GenAI engine powering Hindi conversations, intent detection, agent reasoning, budget planning, and price forecasting
- **Lambda + API Gateway + DynamoDB:** Serverless compute (4 Lambda functions) with auto-scaling to 1000+ users, sub-millisecond database reads across 5 DynamoDB tables, and secure WhatsApp webhook entry point
- **S3 + Secrets Manager + CloudWatch:** Image storage for crop photos, encrypted API key management (zero hardcoded credentials), and real-time monitoring with error/cost alerts

What Value the AI Layer Adds

- Instant **expert advice in 2-5 seconds** in natural Hindi — replaces expensive consultants with a WhatsApp chat that has **zero learning curve**
- **Cross-domain intelligence** in one conversation — ask about a disease and get diagnosis + treatment cost + government scheme eligibility + community alerts to nearby farmers, all in a single response
- **Every response is actionable**, not generic — specific pesticide names, exact dosages, rupee costs, mandi names, and step-by-step next actions personalized to the farmer's crops, land, and location



List of features offered by the solution

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WhatsApp-Based Multi-Agent AI Assistant

Farmer Onboarding – *Gathering Data to Build a Localized Knowledge Graph*

Crop Agent

- Hyper-local crop advice (village level knowledge graph)
- Disease & pest diagnosis (image + text)
- Fertilizer & pesticide recommendations
- Best practices from nearby successful farmers
- Soil & weather aware suggestions based on local conditions

Market Agent

- Live prices from 1000+ mandis, multi-crop comparison
- 7-day AI price forecast + supply-demand signals
- Best sell timing, target price, risk assessment
- Best mandi by price vs distance vs transport cost
- Custom price alerts on target hit or changes

Finance Agent

- Full budget: cost breakdown + revenue + profit projection
- Auto scheme match (PM-KISAN, KCC, PMFBY, SHC)
- Loan advisory: rates, eligibility, docs checklist
- Risk scoring: weather, market, disease probability
- Input cost optimization + expense tracking

Process Flow Diagram

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Onboarding

Farmer sends "Hi" on WhatsApp

AI collects basic details:

- Village name & Location
- Current crop & Sowing date
- Irrigation type & Budget range

System creates Farmer Profile + Village Knowledge Graph

Pre-Planting

"Kaunsi fasal lagau?"

Market Agent:

Demand trends + forecasting → best crop

Finance Agent:

Budget plan + schemes

Crop Agent:

Soil prep + seed variety

"Soybean → 18% higher margin"

Growing

"Agli baarish kab aa rahi hai?"

Disease Diagnosis:

- Detects disease
- Suggests correct pesticide & dosage

Weekly Advisory:

- Weather/Pest alerts
- Irrigation reminders
- Fertilizer schedule by crop stage

Cost Control:

- Compares local market prices
- Cheaper alternatives
- Checks subsidy availability

Harvest

"Kab harvest karu?"

Market Agent:

- Analyzes mandi price trends
- Predicts 2–4 week price movement
- Recommends best nearby mandi

Suggests:

- Harvest now OR wait
- Best nearby mandi

Selling

"Kab harvest karu?"

Market Agent:

- Where to sell (mandi / private buyer)
- Estimated price range
- Expected demand analysis

Finance Agent:

- Final profit calculation
- ROI analysis for the season
- Recommendations for next cycle

Complete cycle: From "Hi" on WhatsApp to profit in bank, powered by 3 AI agents working together

Wireframes/Mock diagrams of the proposed solution

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Kisaan-Friendly User Experience

Simple via WhatsApp
- No app download needed
- Works on basic smartphones
- Image, text & voice support

Local Language First
- Hindi & regional languages
- Easy for rural users
- Voice-friendly interface

Cost & Efficiency First
- Zero Cost Advisory for Farmers
- Low-Latency Responses
- Concise and Actionable Recommendations

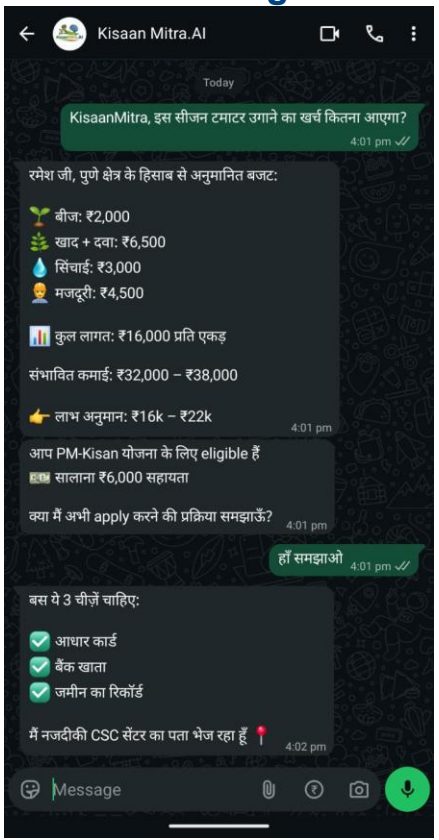
Crop Agent



Market Agent



Finance Agent



Architecture Diagram



AWS Services Used

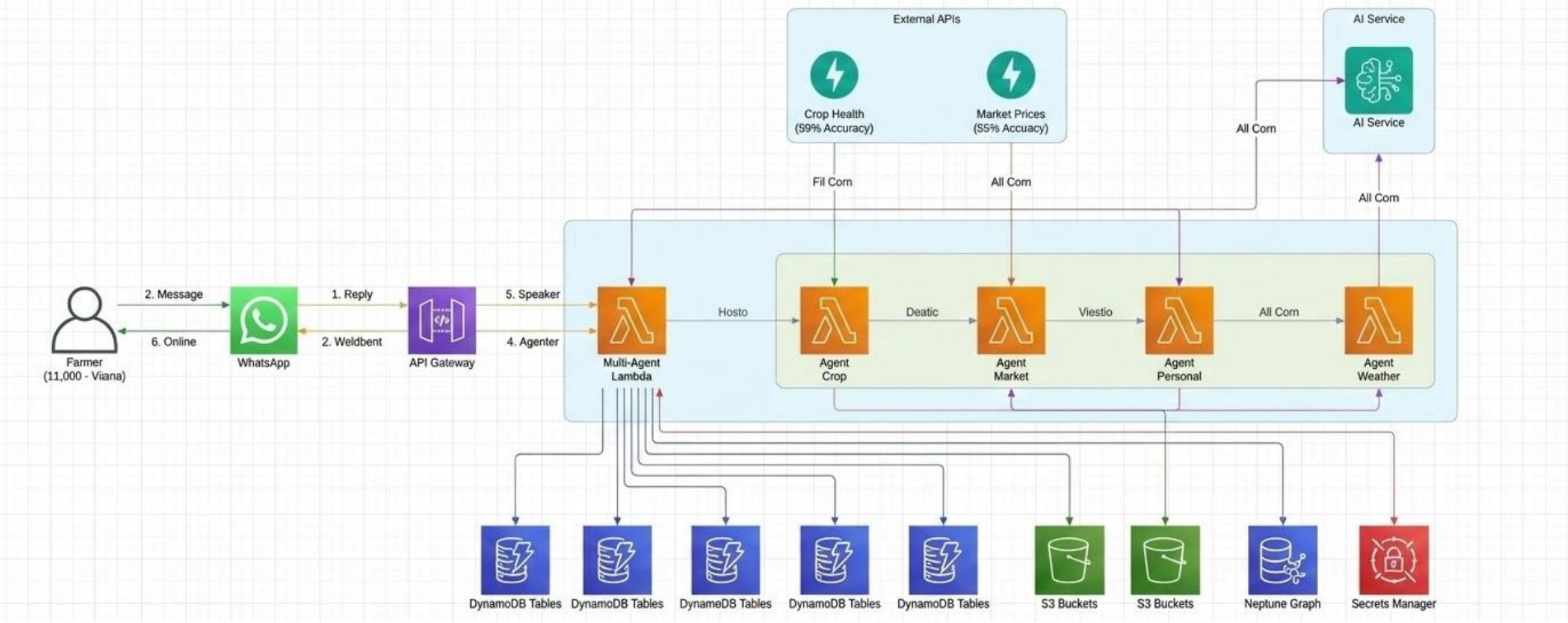
- **Amazon Bedrock** — Nova Pro + Claude Sonnet 4
- **AWS Lambda (x4)** — Orchestrator + 3 Agents
- **Amazon DynamoDB** — 5 tables, sub-ms latency
- **Amazon S3** — Crop images & documents
- **API Gateway** — Secure webhook entry
- **Secrets Manager** — Encrypted API keys
- **CloudWatch** — Logs, metrics, alerts

External APIs

- **Kindwise API** — Disease detection 85%+
- **AgMarkNet API** — Govt mandi prices
- **OpenWeather API** — 7-day forecasts

AWS Architecture Diagram

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Technologies Used in the Prototype:

Interface Layer

WhatsApp Business API
Primary user interface (500M+ users in India)

Streamlit Dashboard
Real-time analytics with interactive visualizations

REST APIs
Webhook integration for message delivery

Backend & Agent Orchestration

AWS Lambda
Serverless compute (Python 3.11, 1024MB, ap-south-1)

LangGraph
Multi-agent orchestration framework (4 agents)

Python 3.11
Core backend language with boto3 SDK

AI & ML Layer

Amazon Bedrock Nova Pro
Primary LLM (100% routing, 92.86% extraction)

LangChain
AI framework for agent coordination & context

Kindwise API
Crop disease detection (99% accuracy)

AgMarkNet + OpenWeather
Govt mandi prices + 7-day weather forecasts

Data & Storage

Amazon DynamoDB
NoSQL database (7 tables, pay-per-request, TTL)

Amazon S3
Object storage (images, PDFs, versioning enabled)

Amazon Neptune (Optional)
Graph database for village knowledge network

AWS API Gateway
Secure webhook entry, auth, rate limiting

Deployment & Scale

AWS CloudWatch
Monitoring, logging, metrics (2.96s avg response)

Amazon EventBridge
Scheduled automation (daily/weekly updates)

AWS Secrets Manager
Secure API key management and rotation

Region: ap-south-1 (Mumbai)
Low-latency serverless for Indian farmers

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Estimated implementation cost:

1 Village Pilot

Tech Infrastructure

AI/LLM API • WhatsApp Business API • Cloud Hosting

₹6–17K/mo

Village Data Agent

1 local agent (₹10–12K/mo) • Training • Smartphone

₹10–12K/mo

Data Collection

Govt API integration • Soil testing kits • Farmer surveys

One-time

Miscellaneous

SIM / Data recharge • Travel • Printing materials

₹3–5K/mo

One-Time Setup

₹28K–40K

Training + Device + Data Setup

Monthly Running Cost

₹19–34K

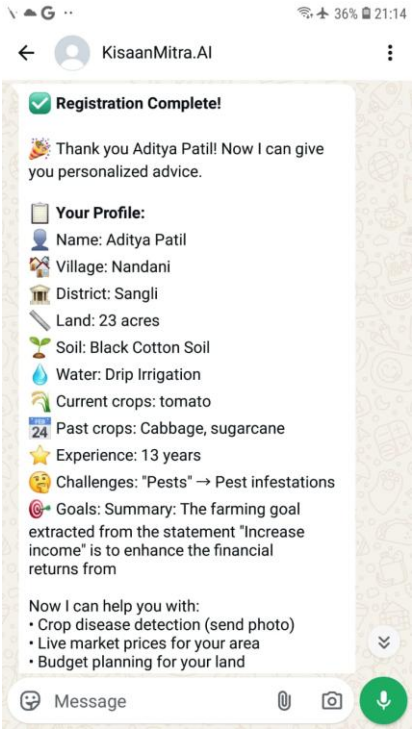
Agent + Tech + Operations

Snapshots of the prototype:

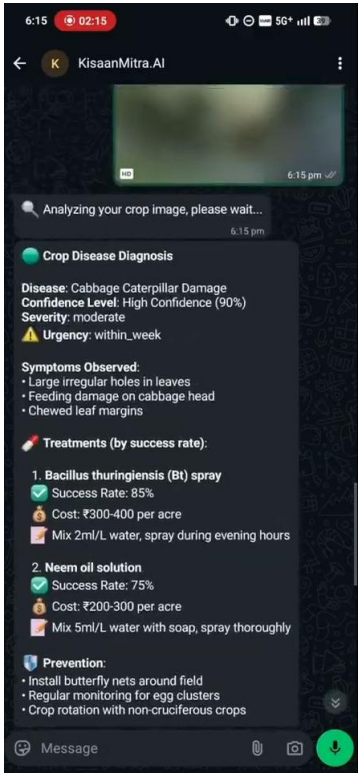
Click here to see the demo video

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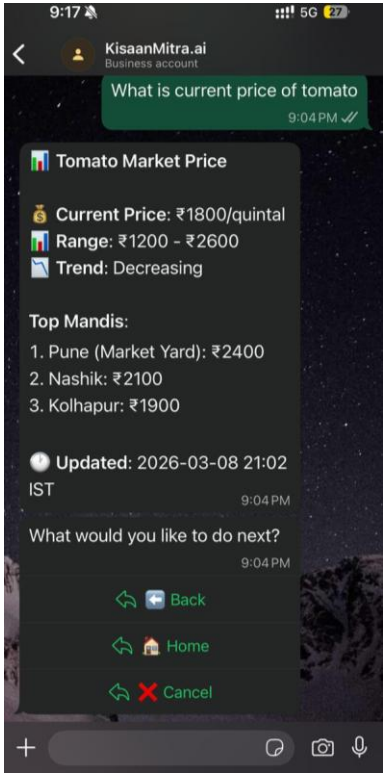
1. Onboarding



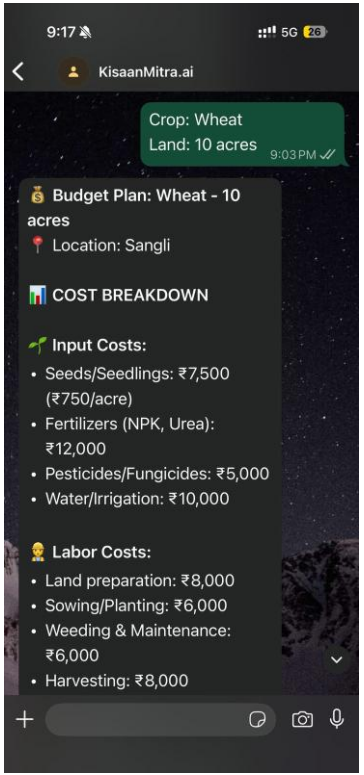
2. Crop Agent



3. Market Agent



4. Finance Agent



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Performance Report & Benchmarking

Verified Metrics (Tested March 7, 2026)

AI Routing Accuracy: 100%

40 queries tested (Hindi + English) with zero misrouting errors across all 4 agents

40 Queries

Crop Name Extraction: 92.86%

13/14 queries correctly extracted crop names from Hindi and English farmer messages

14 Queries

Avg Response Time: 2.96 seconds

Greeting: 1.0s | Market: 2.67s | Finance: 3.73s | Crop: 4.46s

Bilingual Support: 100%

20 Hindi + 20 English queries tested with perfect accuracy in both languages

40 Queries

100% Routing

Verified across Crop, Market, Finance, and General agents

Sub-3s Avg

Latency

Additional Details/Future Development:

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Data Moat

Village-level agricultural data does not exist in India today. Government data stops at district level. Every farmer interaction generates hyper-local data that no competitor can replicate — the longer we operate, the deeper the moat.

Every Recommendation = Revenue

Unlike traditional ad models, our AI agents deliver recommendations at the exact moment of farmer need. Every suggestion is a high-conversion touchpoint for partners — not an interruption, but a solution.

Zero Marginal Cost at Scale

Once AI agents are trained and village data is seeded, adding the next village costs only one field agent (~₹10-12K/month). Tech, AI models, and data pipelines scale at near-zero marginal cost.

Multi-Sided Network Effect

More farmers → richer data → better AI → more partners willing to pay → better services for farmers → more farmers. A self-reinforcing flywheel that is hard to disrupt once established.

"We don't just build an AI tool - we build India's first village-level agricultural data infrastructure."

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Prototype Assets:

- ➔ GitHub Public Repository Link
- ➔ Demo Video Link

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About the Team



Aditya Rane

Advanced Analytics

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Data Science | Ad-Tech | Agentic AI



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Robotics | AI Systems | Computer Vision



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Innovation partner H2S HACK2SKILL | Media partner YOURSTORY

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Thank You

